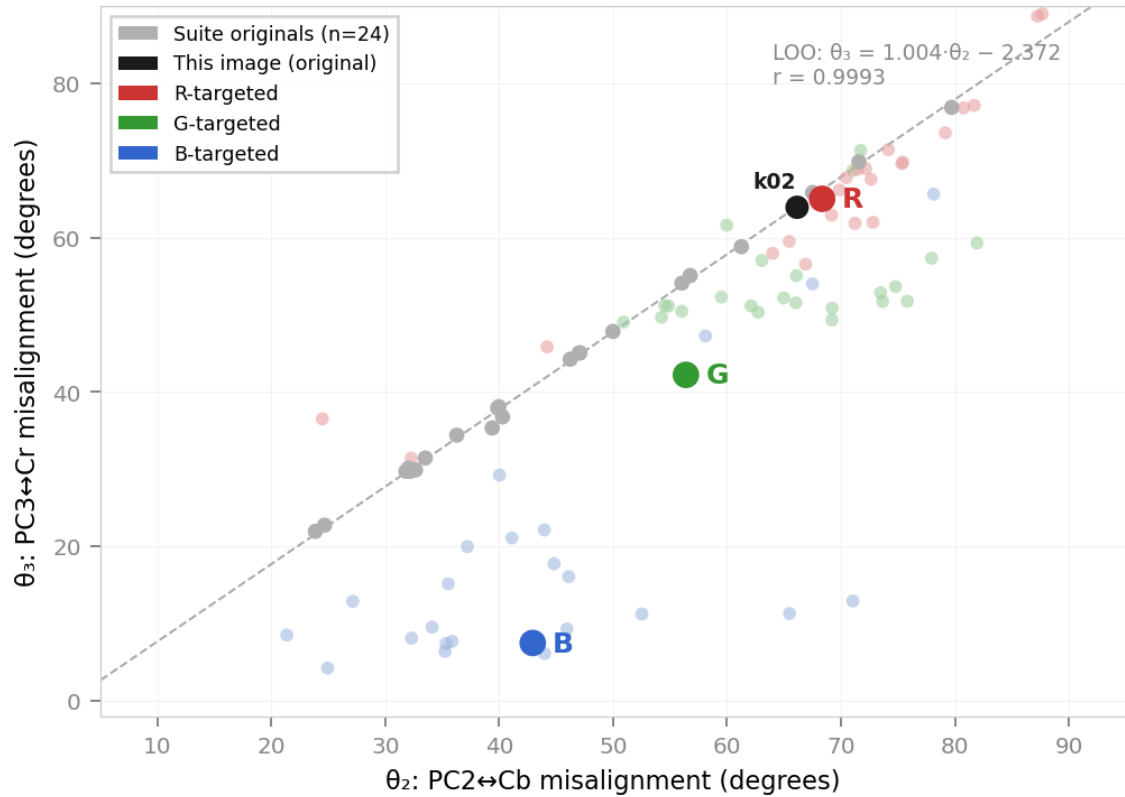


Compression Profile: KODIM02 768x512 | 24-bit RGB | PNG (lossless)

red door | Three-Dimensional (PC1 = 73.37%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.5279	0.4063	0.9740	0.636	66.15°	63.96°	-0.08°
R-targeted	0.1711	0.1399	0.9562	0.422	68.35°	65.06°	-1.19°
G-targeted	-0.0382	0.9879	-0.1739	0.400	56.39°	42.23°	-12.01°
B-targeted	0.9246	0.0640	-0.1132	0.367	42.94°	7.46°	-33.28°

R-targeted isolates red (R-G, R-B < 0.17) while preserving G-B coupling at 0.956.
G-targeted isolates green (R-G, G-B < 0.17) while preserving R-B at 0.988.
B-targeted preserves R-G at 0.925 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM02

RGB | PNG (lossless)

red door | Three-Dimensional (PC1 = 73.37%)

768x512 | 24-bit

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

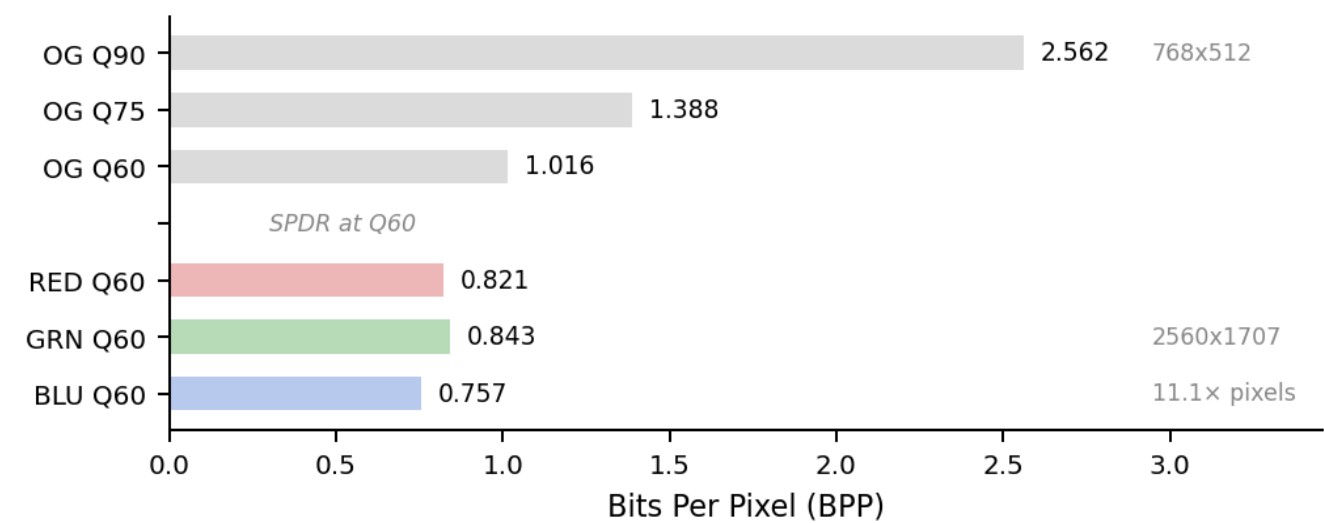
ORIGINAL (kodim02)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	603.5 KB	12.573	393,216
JPEG Q90	JPEG	768x512	123.0 KB	2.562	393,216
JPEG Q75	JPEG	768x512	66.6 KB	1.388	393,216
JPEG Q60	JPEG	768x512	48.8 KB	1.016	393,216

REDISTRIBUTED (kodim02 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	437.8 KB	0.821	4,369,920
GRN Q60	JPEG	2560x1707	449.7 KB	0.843	4,369,920
BLU Q60	JPEG	2560x1707	403.7 KB	0.757	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM02

RGB | PNG (lossless)

red door | Three-Dimensional (PC1 = 73.37%)

768x512 | 24-bit

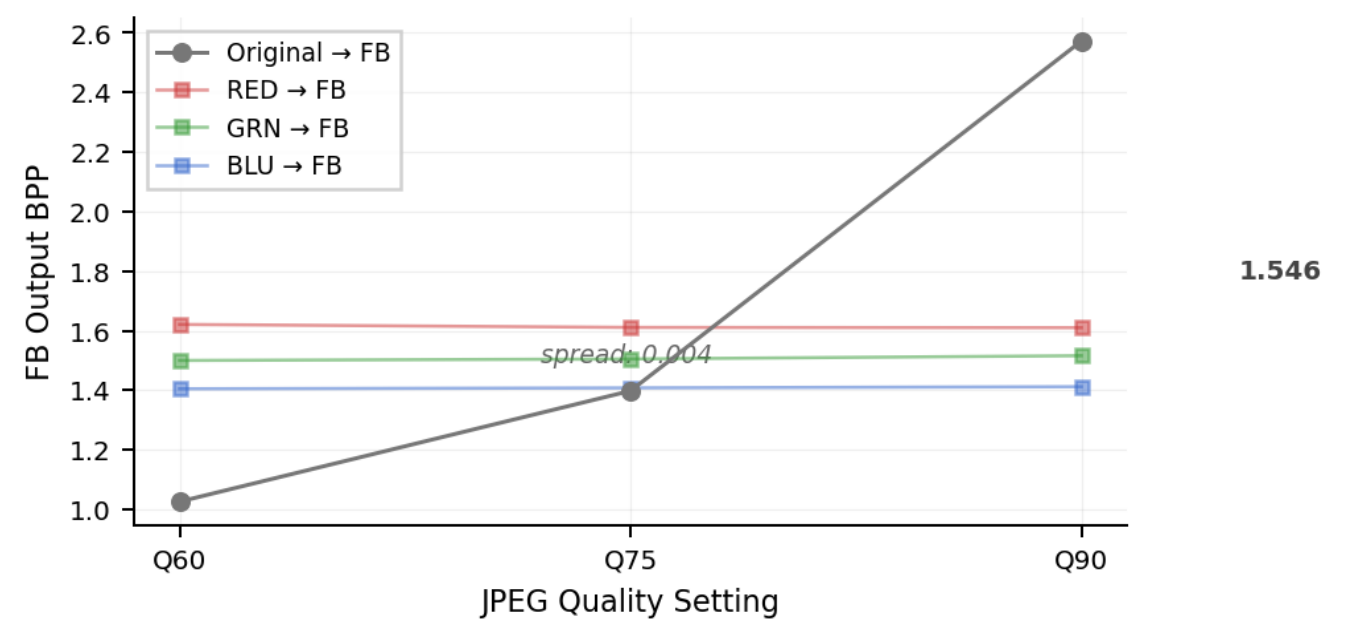
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	2.282	—
R-targeted → FB	2560x1707	2048x1366	1.610	29.4%
G-targeted → FB	2560x1707	2048x1366	1.516	33.6%
B-targeted → FB	2560x1707	2048x1366	1.412	38.1%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.235	45.9%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.6269 vs 0.636 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.546 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.004 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim02

Best BPP (Q60, direct)	0.757	2560x1707, 4,369,920 px
Best BPP through FB	1.412	2048x1366 output
vs OG Q90 → FB	45.1%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216